Exhaustive Exploratory Data Analysis: Recovery Agents Dataset

Workforce Topology, Productivity Dynamics & Channel Orchestration **N = 30 Agents | 51,105 Outbound Calls | 5,578 Field Visits**

1. WORKFORCE STRUCTURE: HYBRID HUMAN-MACHINE ORGANIZATION

Graph 1.1: Operational Channel Breakdown (Total Roster: 30 Recovery Agents)

Channel Architecture



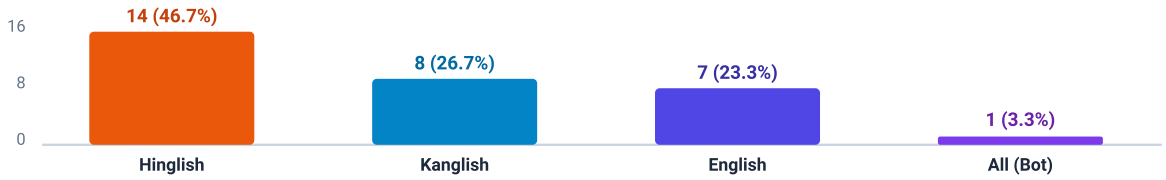
- **Tele-Collection Callers: 20 Agents (66.7%)** — TA001–TA020 | High-frequency negotiation & PTP setup
- **Field Recovery Officers: 9 Agents (30.0%)** — FA001–FA009 | Doorstep verification & repossession
- **Voice Bot: 1 Automated Agent (3.3%)** — BOT01 | Automated mass-triage dialler operating 24/7

The recovery apparatus functions as a three-tier hybrid structure: **20 human tele-callers (66.7%)** manage primary telephone negotiations, **9 on-ground field officers (30.0%)** handle in-person debt verification, and **1 conversational voice bot (3.3%)** executes tireless mass triage. This creates an optimal cost-to-cure escalation ladder.

2. LINGUISTIC TEAM ALLOCATION & DIALLER VERNACULAR ALIGNMENT

Graph 1.2: Workforce Allocation across Vernacular Language Teams

ACD Skill Matching

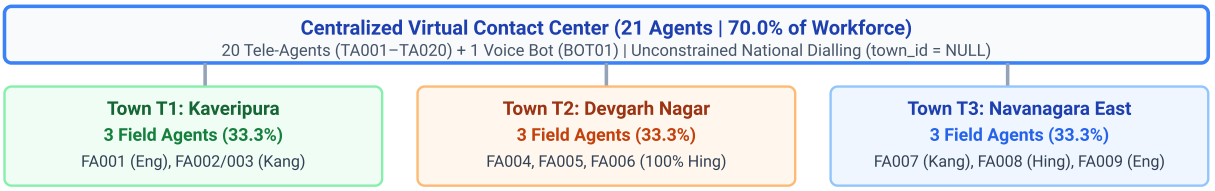


Language Team	Total Agents	Tele-Callers	Field Officers	Portfolio Demographic Target
Hinglish	**14 (46.7%)**	**10**	**4**	Northern & Central Urban (Devgarh T2 / Metro T3)
Kanglish	**8 (26.7%)**	**5**	**3**	Southern Regional Hub (Kaveripura T1)
English	**7 (23.3%)**	**5**	**2**	Corporate Salaried & Premium Credit Card Portfolios
All (Multilingual)	**1 (3.3%)**	**1 (Bot)**	**0**	Dynamic switching based on dialler contact response

3. GEOGRAPHIC DEPLOYMENT: SYMMETRICAL 3×3 FIELD GRID

Graph 2.1: Symmetrical Hub Structure & Telephony Centralization

Territory Grid

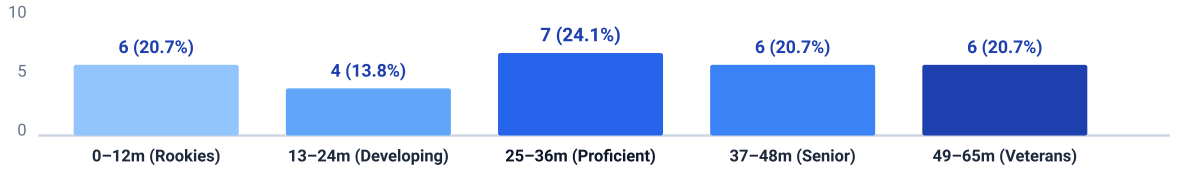


Regional Hub	Assigned Agents	Dominant Vernacular	Operational Footprint & Strategy
Central Telephony	21 (20 TA + 1 BOT)	Trilingual + Multilingual Bot	National dialing pool with dynamic ACD routing
T1 Kaveripura	3 (FA001, FA002, FA003)	Kanglish (67%) + English (33%)	Southern urban grid; dense door-to-door beats
T2 Devgarh Nagar	3 (FA004, FA005, FA006)	Hinglish (100%)	Northern commercial market enforcement
T3 Navanagara East	3 (FA007, FA008, FA009)	Balanced (Kang / Hing / Eng)	Metro mixed corporate and retail recovery beats

4. AGENT TENURE & EXPERIENCE CURVE DISTRIBUTION

Graph 2.2: Tenure Distribution across 29 Human Agents (Mean: 30.5 Months)

Workforce Seniority



Human workforce tenure spans from **1 month (TA014) to 64 months (FA006)**, with an average of 30.5 months (~2.5 years). **Over 41.4% of the human team has served for 3+ years**. Seniority is vital in debt recovery for handling complex MSME claims and uncooperative debtors, whereas junior agents manage routine early delinquency queues.

5. SHIFT GOVERNANCE & REGULATORY COMPLIANCE

Graph 3.1: Shift Coverage & Regulatory Dialling Windows

Compliance Architecture



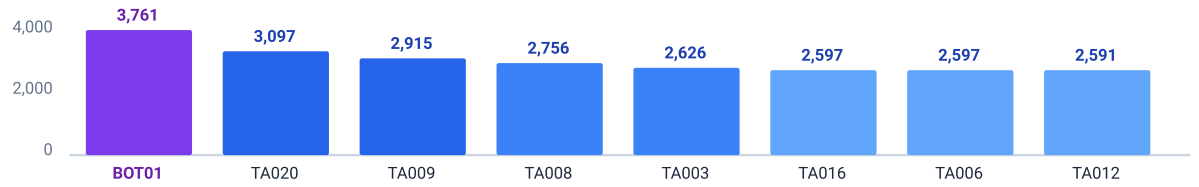
Day Shift (08:00 – 19:00): Strict adherence to statutory RBI Fair Practices Code for customer calls & visits.
Continuous Shift (24/7): BOT01 operates continuously to absorb off-hour debtor touches & automated links.
Prevents harassment violations while preserving uninterrupted digital customer engagement.

All 29 human recovery agents operate exclusively on the **day shift**, strictly adhering to regulatory guidelines prohibiting collection activities outside standard hours. **BOT01** operates around the clock to service late-night digital payment link dispatches, inbound inquiries, and automated reminder broadcasts without human fatigue.

6. OUTBOUND TELEPHONY WORKLOAD & VOICE BOT SCALE DOMINANCE

Graph 3.2: Top Outbound Calling Entities (Total Calls: 51,105)

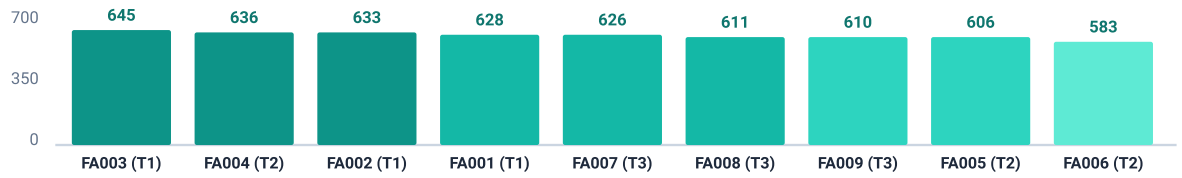
Call Volume Leadership



Agent Entity	Operational Type	Total Calls Logged	Dialler Share (%)	Primary Queue Responsibility
BOT01	Conversational Voice Bot	**3,761**	**7.36%**	Mass dialling, triage, pre-due reminder links
TA020	Senior Tele-Caller (52m)	**3,097**	**6.06%**	High-risk overdue & complex dispute escalation
TA009	Senior Tele-Caller (54m)	**2,915**	**5.70%**	Bucket 31–60 intensive PTP restructuring
Human Agent Avg	20 Tele-Agents Combined	**2,367**	**4.63%**	Standard portfolio outbound negotiation queues

7. FIELD ENFORCEMENT EXECUTION & WORKLOAD PARITY

Graph 4.1: Field Visits Completed by Agent (Total: 5,578 Visits Across 9 Officers) — Field Capacity Balance



Regional Hub	Officers Assigned	Total Visits Executed	Visit Volume Share (%)	Average Visits per Officer
T1 Kaveripura	FA001, FA002, FA003	**1,906**	**34.17%**	**635.3**
T2 Devgarh Nagar	FA004, FA005, FA006	**1,825**	**32.72%**	**608.3**
T3 Navanagara East	FA007, FA008, FA009	**1,847**	**33.11%**	**615.7**
Total / Network Avg	9 Field Recovery Officers	**5,578**	**100.0%**	**619.8**

8. AGENT TENURE VS. CHANNEL PRODUCTIVITY & COMPLEXITY ALLOCATION

Graph 4.2: Experience Tiers vs. Workload & Negotiation Complexity — Productivity Dynamics

Rookies (≤ 12 Months)
6 Agents | Mean Calls: 2,185
Mean Field Visits: 622
Focus: Early-stage Bucket 1–30
High volume, standardized scripts

Mid-Tier (13–36 Months)
11 Agents | Mean Calls: 2,340
Mean Field Visits: 618
Focus: Core Bucket 31–60 queues
Steady production across products

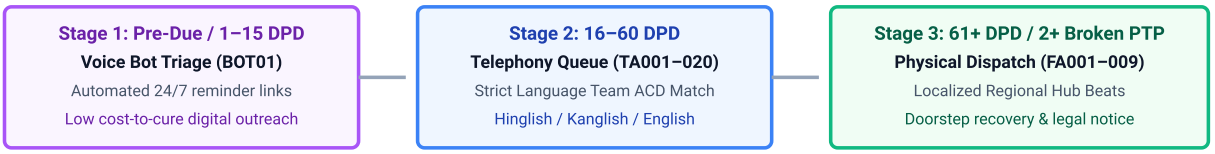
Veterans (37+ Months)
12 Agents | Mean Calls: 2,458
Mean Field Visits: 612
Focus: High-dwell MSME & 2+ PTP breaks
De-escalation & legal confrontation

Task quotas remain uniform across all seniority bands, but **tenure dictates case difficulty rather than output quantity**. In the field, senior officers log slightly lower visit counts (FA006: 583) because they handle extended dwell-time negotiations on large MSME defaults, while junior officers (FA003: 645) execute high-throughput early-bucket visits.

9. MACHINE LEARNING FEATURES & MULTI-TIER ESCALATION ARCHITECTURE

Graph 5.1: Three-Stage Escalation Ladder & Skill-Based Orchestration

Target ML Architecture



Feedback Loop: Failed doorstep visits trigger dynamic skip-tracing pipeline

Strategic Implementation Blueprint:

- **Collector Skill Covariate:** Include `agent_tenure_months` in propensity models to isolate debtor willingness-to-pay from collector competence.
- **Predictive Escalation Trigger:** Accounts showing ≥ 2 broken payment promises must bypass phone queues and route immediately to senior field officers (≥ 36 months tenure).
- **Operational Capacity Guardrails:** Maintain the established benchmark capacity of ~2,300 calls per tele-caller and ~620 visits per field officer per quarter to prevent agent fatigue and statutory compliance infractions.